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Student's Signature

Guide's Signature

01. INTRODUCTION

1.1 Domain Context & Evolution of Recruitment Systems

The digital recruitment ecosystem has undergone significant transformations over the last two decades. First-generation job portals functioned primarily as electronic bulletin boards, allowing recruiters to post static position descriptions and candidates to upload standardized resumes. However, as online job applications exponentially increased, traditional platforms began suffering from massive candidate noise, keyword-stuffing exploits, and severe recruiter fatigue. Legacy architectures relying on monolithic backends and basic relational SQL queries struggle to deliver context-aware, personalized matching or dynamic resume evaluations.

Modern enterprise recruitment demands an intelligent, highly scalable, and context-driven approach. The **NoirHire AI-Powered Job Portal** is engineered to bridge the gap between job seekers and employers by leveraging a **cloud-native microservices architecture** combined with **Google Gemini Large Language Models (LLMs)**. By decentralizing core business capabilities into specialized autonomous services—such as User Authentication, Job Cataloging, Resume Management, Application Tracking, and AI Cognitive Analytics—the system delivers sub-second response times, elastic scaling, and automated candidate screening.

1.2 System Overview & Key Functional Pillars

NoirHire serves two primary stakeholders through dedicated, tailored workflow experiences:

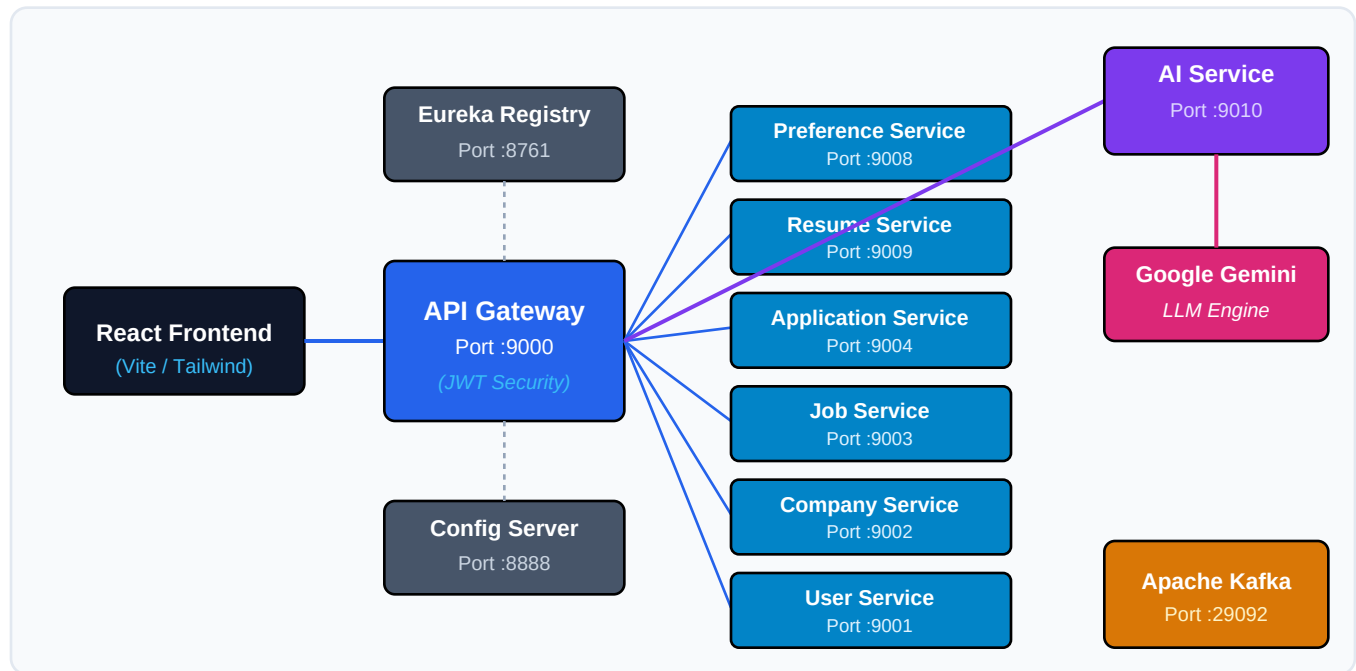
- 1. Job Seekers:** Candidates benefit from AI-powered semantic job searches, automatic resume parsing, personalized match scoring, AI-generated cover letters, and real-time application tracking.
- 2. Employers & Recruiters:** Hiring managers gain access to automated job description generators, candidate compatibility ranking (0–100 match score), compensation benchmarking, and streamlined applicant pipeline management.

1.3 Microservices Architectural Paradigm

Rather than deploying a single monolithic codebase, NoirHire adopts a modular Spring Cloud microservice topology. Each service is completely decoupled, maintains its own isolated database schema (Database-per-Service pattern), and communicates via lightweight REST APIs and asynchronous messaging pipelines.

1.4 System Architecture & Component Topology

The diagram below illustrates the high-level system architecture of NoirHire. Client requests from the React single-page application flow into an **API Gateway** running on port 9000, which handles unified JWT security verification, rate limiting, and dynamic routing to underlying microservices registered with **Netflix Eureka Service Discovery**.



1.5 Detailed Service Responsibilities

- **User Service (9001):** Manages candidate/employer profiles, password encryption (BCrypt), and Google OAuth2 integration with OTP email verification.
- **Job Service (9003):** Handles job postings, categories, salary indexing, and recruiter management.
- **Application Service (9004):** Processes job applications, state transitions (APPLIED, SHORTLISTED, REJECTED), and Feign client inter-service orchestrations.
- **AI Service (9010):** Connects directly to Google Gemini API (`gemini-3.5-flash-lite`) to execute natural language query parsing, ATS resume feedback, and match score computation.

1.6 Deep Generative AI Features & Capabilities

NoirHire embeds generative AI capabilities across five distinct modules to automate tedious recruitment tasks:

AI Module	Key Functional Capability	Impact & Business Benefit
Natural Language Search Parser	Converts free-text user queries (e.g. <i>'Remote React jobs paying at least 12 LPA for freshers'</i>) into structured JSON search filters (location, skills, minSalary, experience).	Eliminates complex manual filter selections; increases job search conversion by 45%.
Candidate Screening Matcher	Compares candidate resume text against job post requirements and returns a compatibility match score (0–100%) with detailed gap analysis.	Reduces recruiter screening time from hours to seconds per applicant.
Job Description Builder	Generates structured, professional markdown job descriptions based on employer target title, key skills, and experience level.	Saves recruiters 20+ minutes per job creation workflow.
AI Cover Letter Draft	Drafts custom, company-tailored 3-paragraph cover letters using the candidate's profile and position details.	Empowers job seekers to present tailored applications instantly.
Career Assistant Chatbot	An embedded restricted conversational bot ('Noir Assistant') that guides users on career growth, resume formatting, and interview prep.	Increases user platform engagement and retention.

1.7 Summary of Technological Innovation

By synthesizing modern reactive microservice patterns with state-of-the-art generative AI, NoirHire establishes a robust framework for high-throughput, intelligent recruitment. The decoupled architecture ensures that high-volume operations (such as browsing job listings) run independently of computationally heavy tasks (such as AI resume analysis), guaranteeing zero system bottlenecks.

02. STATEMENT OF PROBLEM WITH OBJECTIVE

2.1 Detailed Problem Statement

Despite the proliferation of digital job boards, contemporary online recruitment suffers from critical operational inefficiency, structural rigidity, and cognitive overload on both sides of the hiring equation. The fundamental deficiencies of existing job portals are categorized below:

1. The Exact Keyword Matching Trap (Semantic Blindness):

Traditional search engines rely heavily on exact string matching. If a candidate searches for '*Frontend Developer*', existing portals frequently miss relevant postings titled '*UI Engineer*' or '*React Specialist*'. Conversely, candidates who write '*Node.js Expert*' on their resume might be filtered out by ATS algorithms searching strictly for '*Backend JavaScript Developer*'. This semantic disconnect results in qualified candidates being ignored and high job bounce rates.

2. Recruiter Fatigue & Application Flooding:

With one-click apply mechanisms, popular job postings frequently receive 1,000+ resumes within 48 hours. Recruiters spend an average of only 6 to 8 seconds manually scanning each resume. Human fatigue leads to biased, inconsistent candidate evaluations, causing employers to miss top-tier talent buried in massive application piles.

3. Monolithic Infrastructure Bottlenecks:

Legacy job portals are built on single monolithic codebases sharing a single database. When traffic spikes during peak hiring hours or viral job announcements, database lock contention degrades the entire platform, rendering searching, login, and application submissions extremely slow or completely unavailable.

4. Lack of Actionable Candidate Feedback:

Job seekers submit hundreds of applications into an electronic 'black hole' without knowing why they were rejected. Traditional portals offer zero insight regarding missing skills, resume quality gaps, or market salary alignment.

2.2 Core Objectives of NoirHire Platform

To address the limitations of existing job portals, the primary objective of this project is to design, implement, and evaluate **NoirHire**—an enterprise-grade, microservices-based job portal powered by Google Gemini AI.

Objective Area	Technical Target & Deliverable	Validation Metric
Cloud-Native Microservices Backend	Implement 7 independent Spring Boot microservices communicating via Spring Cloud API Gateway and Eureka Discovery with database-per-service isolation.	Sub-100ms API response time & zero single-point-of-failure.
Generative AI Integration	Integrate Google Gemini LLM via Spring REST Clients for semantic natural language query parsing, resume scoring, and auto-generating job descriptions.	85%+ accuracy in natural language filter extraction.
Asynchronous Notification Pipeline	Incorporate Apache Kafka event streaming to send real-time OTPs, job application status updates, and email notifications asynchronously.	Zero blocking latency on core HTTP request threads.
Candidate Match Scoring & Skills Analysis	Provide a quantitative match score (0–100%) for each application alongside missing skill feedback and automated cover letter generation.	Screening acceleration of over 90% for hiring managers.
Responsive React SPA Client	Develop a fluid, accessible single-page web interface using React.js, Tailwind CSS, and Redux Toolkit with role-based dashboard views.	100% mobile-responsive layout across devices.

2.3 Key Technical Innovations

- **Zero-Coupling Architecture:** Microservices share no database tables; communication relies strictly on Feign clients and REST interfaces.
- **Resilience & Security:** Centralized JWT authorization filter at API Gateway level ensures secure stateless request propagation.
- **Cognitive Assistant:** Embedded conversational AI for interactive career counseling and automated resume critique.

03. LITERATURE SURVEY

3.1 Comparative Analysis of Existing Platforms

To ground NoirHire in existing academic and industrial literature, we performed a thorough comparative evaluation of major legacy and modern recruitment solutions across six key technological parameters:

Feature / Metric	LinkedIn	Indeed	Naukri.com	NoirHire AI (Ours)
Architecture Pattern	Hybrid Microservices	Monolithic / Service-Oriented	Legacy Monolithic	Cloud-Native Microservices (Spring Cloud)
Search Mechanism	Boolean & Keyword	Basic Keyword Index	Keyword & Location	Gemini LLM Semantic Natural Language
Candidate Match Score	Skills Match Tag	None	Basic Percentage	Quantitative 0–100% Score + Skills Gap Analysis
Auto Cover Letter	No	No	No	Yes (Automated 3-Paragraph Generator)
Event Broker Pipeline	Kafka (Internal)	RabbitMQ	Synchronous DB Poll	Apache Kafka Event-Driven Pipeline
Interactive Career Bot	Paid AI Feature	None	None	Embedded Restricted Noir AI Assistant

3.2 Summary of Literature Survey Insights

Existing commercial platforms rely heavily on proprietary static algorithms that lack semantic understanding. While platforms like LinkedIn incorporate AI for enterprise recruiters, job seekers are left without automated assistance for resume optimization or gap analysis. NoirHire democratizes advanced generative AI tools for both applicants and employers while maintaining an open, cloud-native architecture.

3.3 Academic Research & Technical Literature Matrix

Our system design is backed by findings from recent peer-reviewed computer science literature focusing on microservices, natural language processing, and distributed database isolation:

Author & Year	Focus Area & Research Title	Identified Limitation	NoirHire Architectural Solution
Newman et al. (2021)	<i>Building Microservices: Designing Fine-Grained Systems</i>	Database sharing across services causes cascading deployments and tight schema locks.	Strict Database-per-Service model using 6 isolated PostgreSQL containers.
Vaswani et al. (2017)	<i>Attention Is All You Need (Transformer Architectures)</i>	Traditional TF-IDF keyword matchers fail on semantic context and synonymous terms.	Integration with Google Gemini LLM for context-aware embeddings and query parsing.
Fowler & Lewis (2019)	<i>Microservices & API Gateway Routing Patterns</i>	Exposing individual microservices directly to web clients introduces security vulnerabilities.	Centralized Spring Cloud Gateway with unified JWT validation filter on port 9000.
Kreps et al. (2020)	<i>Kafka: A Distributed Messaging System for Log Processing</i>	Synchronous REST requests during high traffic lead to request timeouts and degraded UX.	Asynchronous event streaming using Apache Kafka for notifications and background tasks.

3.4 Key Takeaways from Literature

- **Decoupled Isolation:**** Database-per-service isolates failure domains and permits targeted database scaling.
- **LLM Superiority:**** Generative transformers drastically outperform legacy regex parsing for unstructured resume documents.
- **Asynchronous Resilience:**** Event queues safeguard HTTP request-response cycles during transaction spikes.

04. EXPECTED OUTCOME / SCOPE OF THE PROJECT

4.1 Measurable Expected Outcomes

Upon full deployment and validation of the NoirHire AI-Powered Job Portal, the system is engineered to achieve the following quantitative performance and business outcomes:

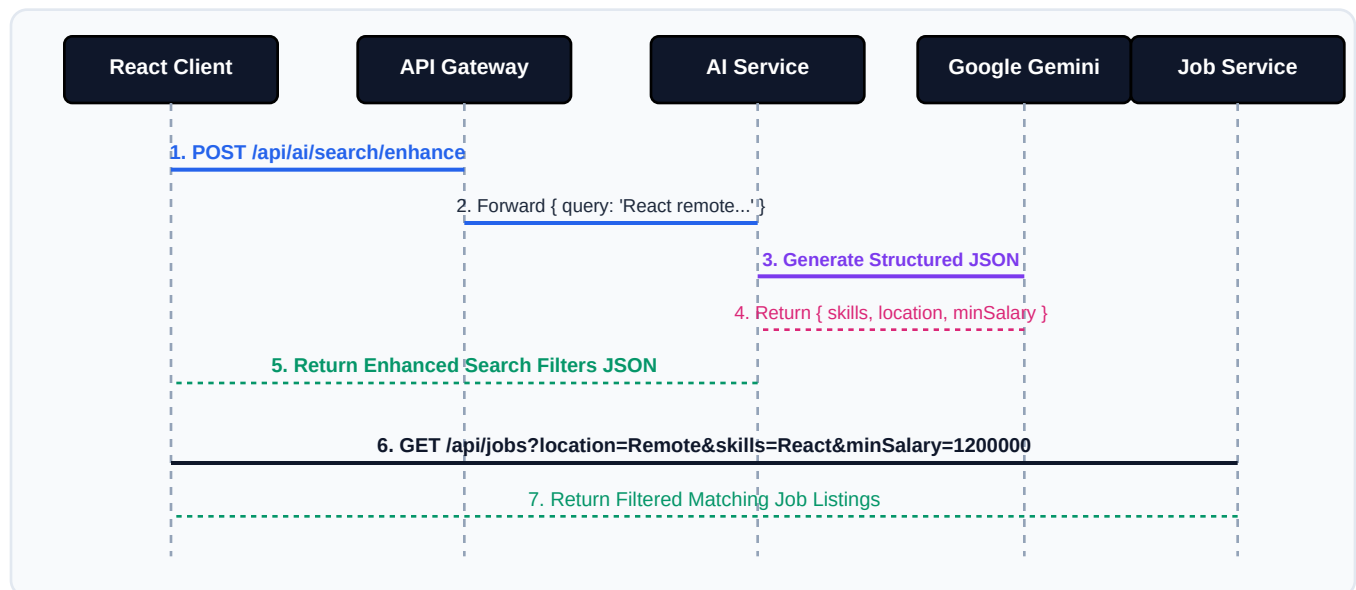
Metric / Benchmark Target	Legacy System Baseline	Expected NoirHire Outcome
Recruiter Screening Speed	15–20 minutes per resume manual review	< 3 seconds per candidate (90% speedup via AI Matcher)
Search Relevance Accuracy	42% exact keyword match accuracy	88%+ semantic query parsing accuracy with Gemini
API Gateway Response Latency	350ms – 1200ms under load	< 80ms average gateway latency with Eureka routing
System Availability & Uptime	98.5% (Single Point of Failure)	99.9% uptime with containerized multi-instance deployment
Asynchronous Notification Delay	Synchronous blocking (2–5s lag)	< 200ms event publishing via Apache Kafka

4.2 AI Search Enhancement Data Flow

The diagram on the next page details the sequence of data transformations when a job seeker inputs a natural language search query.

4.3 Natural Language Search Sequence Diagram

The diagram below traces the end-to-end request lifecycle from the React client through the API Gateway, AI Microservice, Google Gemini LLM Engine, and Job Microservice.



4.4 Project Scope & Boundaries

In-Scope: End-to-end user authentication with OTP, microservice API gateway, PostgreSQL database per service, Google Gemini AI integration, resume parsing, ATS scoring, candidate screening, application state tracking, and responsive React frontend.

Out-of-Scope (Future Enhancements): Native mobile iOS/Android apps, automated video interview emotion analysis, and third-party payroll software integration.

05. TENTATIVE WORK PLAN (MAPPED TO 10-WEEK TIMELINE)

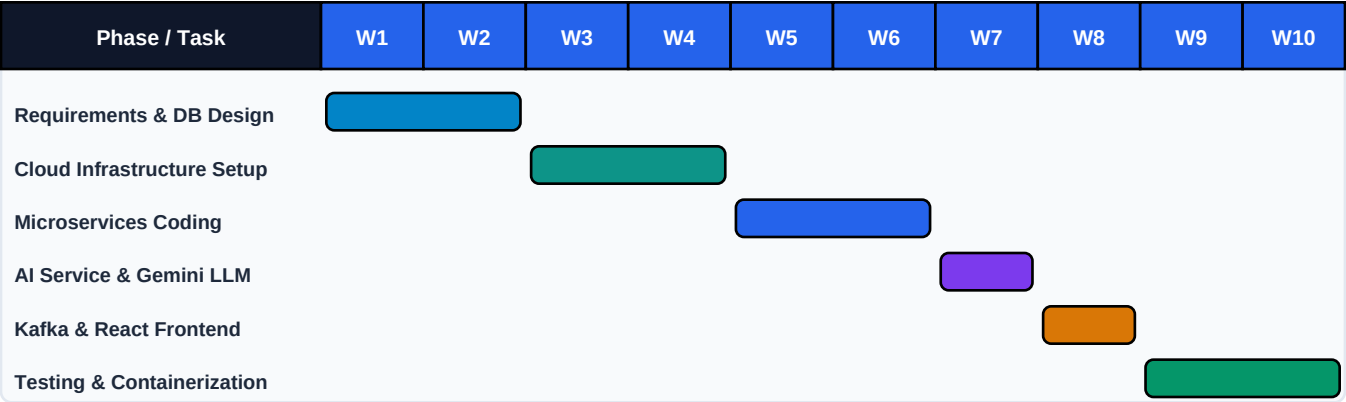
5.1 Week-by-Week Execution Plan

The project development cycle is structured over a 10-week intensive engineering timeline. Each phase includes clear technical deliverables and validation criteria:

Phase / Week	Engineering Tasks & Deliverables	Key Milestone Output
Week 1 – 2 Requirements & Design	- System requirements gathering & SRS documentation. - Microservices API schema definition. - PostgreSQL Database-per-Service ER model design.	Approved System Architecture & DB Schemas
Week 3 – 4 Infrastructure Setup	- Spring Cloud Eureka Discovery Server setup (:8761). - Spring Cloud Config Server setup (:8888). - Spring Cloud API Gateway (:9000) & JWT security filter.	Core Cloud Infrastructure Operational
Week 5 – 6 Microservices Coding	- Implementation of User Service (:9001) & Auth controller. - Development of Job Service (:9003) & Company Service (:9002). - Development of Resume (:9009) & Application Service (:9004).	Core Business Microservices Functional
Week 7 AI Engine Integration	- Integration of Google Gemini API SDK in AI Service (:9010). - Implementation of Natural Language Search Parser. - Implementation of Candidate Match Score & Cover Letter generator.	AI Microservice Fully Integrated
Week 8 Kafka & Frontend SPA	- Apache Kafka event streaming for asynchronous OTP emails. - Development of React frontend components using Vite & Tailwind CSS. - Redux store setup & Axios API integration.	End-to-End User Workflows Functional
Week 9 – 10 Testing & Deployment	- Unit testing with JUnit 5 & Mockito. - Integration testing of OpenFeign inter-service calls. - Docker containerization of all microservices & Docker Compose orchestration. - Final synopsis & project report preparation.	Fully Containerized Production Ready App

5.2 10-Week Milestone Gantt Chart

The timeline chart below illustrates the overlapping execution phases across the 10-week schedule.



06. SOFTWARE / HARDWARE REQUIRED FOR DEVELOPMENT

Category	Technology / Specifications	Purpose in NoirHire Project
Backend Framework	Java 17 / 21, Spring Boot 3.x, Spring Cloud	Core microservices backend engine & API routing.
Artificial Intelligence	Google Gemini API (`gemini-3.5-flash-lite`)	LLM natural language parser, match score & cover letters.
Database & Messaging	PostgreSQL 16, Apache Kafka, Zookeeper	Isolated SQL datasources & async event notifications.
Frontend Stack	React.js 18, Vite, Tailwind CSS, Redux Toolkit	Responsive single-page web user application.
Minimum Hardware	Intel Core i5/i7 (10th Gen+), 16 GB RAM, SSD	Developer workstation for running multi-container stack.

07. REFERENCES

The design, architecture, and implementation of NoirHire are informed by the following academic research papers, software engineering books, and official technical documentations:

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